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COURSE CODE: SBT-DF203
LAB NAME: WEP40 WIRELESS PACKET DECRYPTION AND AIRCRACK FORENSICS
DATE: 23/09/2026
INSTRUCTOR: AMINU IDRIS

1. Source Materials and Slide Alignment

- Decrypt Wireless Packets: IEEE 802.11 frames, WEP/RC4, IV reuse, Aircrack-ng key recovery, airdecap-ng decryption, file carving and optional passphrase analysis.
- The slide uses the historical CodeGate CTF 2015 Good Crypto capture and the known WEP40 key A4:3D:F6:F3:74.
- The assignment asks for host/server IP addresses, MAC addresses, manually extracted images/HTML and the router password where the optional CTF challenge is completed.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

3. Learning Outcomes

- Identify management, control and data frames in an 802.11 capture.
- Explain WEP40 structure, RC4 keystream use, IV reuse and integrity limitations.
- Use Aircrack-ng on a supplied historical capture only.
- Decrypt WEP traffic using airdecap-ng and verify output files.
- Extract IP/MAC endpoints, protocols, images and HTML from decrypted traffic.
- Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
- Recommend modern wireless security controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.

- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Do not capture, crack or connect to any live wireless network. This lab is exclusively offline and limited to the supplied historical training file.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
```

```
cd ~/SBT-DF203-Lab9
```

```
pwd
```

```
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
```

```
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
```

```
wget -O evidence/file.xz \
```

```
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
```

```
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
```

```
cp --preserve=timestamps evidence/file.xz working/file_working.xz
```

```
unxz -k working/file_working.xz
```

file working/file_working

sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt

Screenshot required: Compressed evidence, decompressed capture type and SHA-256 hashes.

ANSWER:

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwva
  Metasploitable 2
  Kali Linux

Kali Linux
collins@kali: ~/SBT-DF203-Lab9

Session Actions Edit View Help
(collins@kali)~[~/Desktop]
$ mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
(collins@kali)~[~/Desktop]
$ cd ~/SBT-DF203-Lab9
(collins@kali)~[~/SBT-DF203-Lab9]
$ pwd
/home/collins/SBT-DF203-Lab9
(collins@kali)~[~/SBT-DF203-Lab9]
$ find . -maxdepth 1 -type d -print
.
./evidence
./screenshots
./scripts
./exported
./reports
./working

(collins@kali)~[~/SBT-DF203-Lab9]
$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
[sudo] password for collins:
aircrack-ng is already the newest version (1:1.7+git20230807.4bf83f1a-3).
aircrack-ng set to manually installed.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
xz-utils is already the newest version (5.8.3-1).
The following packages were automatically installed and are no longer required:
  libpython3.13-dev python3.13 python3.13-dev python3.13-minimal python3.13-tk python3.13-venv rpcsvc-proto
Use 'sudo apt autoremove' to remove them.

Upgrading:
  binwalk python3-binwalk
Installing:
  foremost

Summary:
Upgrading: 2, Installing: 1, Removing: 0, Not Upgrading: 1276
Download size: 170 kB
Space needed: 105 kB / 59.0 GB available

Ign:1 https://http.kali.org/kali kali-rolling/main amd64 python3-binwalk all 2.4.3+dfsg1-3
Ign:2 https://http.kali.org/kali kali-rolling/main amd64 binwalk all 2.4.3+dfsg1-3
Ign:3 https://http.kali.org/kali kali-rolling/main amd64 foremost amd64 1.5.7-12+b1
Ign:1 https://http.kali.org/kali kali-rolling/main amd64 python3-binwalk all 2.4.3+dfsg1-3
Ign:2 https://http.kali.org/kali kali-rolling/main amd64 binwalk all 2.4.3+dfsg1-3
Ign:3 https://http.kali.org/kali kali-rolling/main amd64 foremost amd64 1.5.7-12+b1
Get:1 https://http.kali.org/kali kali-rolling/main amd64 python3-binwalk all 2.4.3+dfsg1-3 [122 kB]
Get:2 https://http.kali.org/kali kali-rolling/main amd64 binwalk all 2.4.3+dfsg1-3 [6,008 B]
Get:3 https://http.kali.org/kali kali-rolling/main amd64 foremost amd64 1.5.7-12+b1 [41.8 kB]
Fetched 170 kB in 3min 28s (817 B/s)
(Reading database... 426274 files and directories currently installed.)
Preparing to unpack .../python3-binwalk_2.4.3+dfsg1-3.all.deb...
Unpacking python3-binwalk (2.4.3+dfsg1-3) over (2.4.3+dfsg1-2)...
Preparing to unpack .../binwalk_2.4.3+dfsg1-3.all.deb...
Unpacking binwalk (2.4.3+dfsg1-3) over (2.4.3+dfsg1-2)...
Selecting previously unselected package foremost.
Preparing to unpack .../foremost_1.5.7-12+b1_amd64.deb...
Unpacking foremost (1.5.7-12+b1)...
Setting up python3-binwalk (2.4.3+dfsg1-3)...
Setting up binwalk (2.4.3+dfsg1-3)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...

(collins@kali)~[~/SBT-DF203-Lab9]
$ wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
Prepared http:// to 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypt
--2026-09-23 11:35:12-- http://%20/
Resolving ()... failed: Name or service not known.
wget: unable to resolve host address 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypt
--2026-09-23 11:35:12-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypt
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz      100%[=====] 12.24M  131KB/s  in 53s

2026-09-23 11:36:40 (238 KB/s) - 'evidence/file.xz' saved [12837660/12837660]

FINISHED --2026-09-23 11:36:40--
Total wall clock time: 1m 28s
Downloaded: 1 files, 12M in 53s (238 KB/s)
Downloaded: 1 files, 12M in 53s (238 KB/s)

(collins@kali)~[~/SBT-DF203-Lab9]
$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz

(collins@kali)~[~/SBT-DF203-Lab9]
$ cp --preserve=timestamps evidence/file.xz working/file_working.xz

(collins@kali)~[~/SBT-DF203-Lab9]
$ unxz -k working/file_working.xz

(collins@kali)~[~/SBT-DF203-Lab9]
$ file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(collins@kali)~[~/SBT-DF203-Lab9]
$ sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working

(collins@kali)~[~/SBT-DF203-Lab9]
$
```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab9-COLLINS KELECHI AMADI
Trainee name	COLLINS AMADI
Date and time started	23-09-2026
Evidence file name(s)	evidence/file.xz tee reports/file_xz_sha256.txt
Source or generation method	COLLINS@KALI
Original SHA-256	zdd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working-copy SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
Analysis workstation/VM	KALI LINUX
Notes on any changes	NILL

8. Part A - Inventory the Wireless Capture

CAP=working/file_working

capinfos "\$CAP" | tee reports/capinfos.txt

tshark -r "\$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

tshark -r "\$CAP" -Y 'wlan' -T fields \

-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \

| head -n 100 | tee reports/wlan_frame_sample.tsv

Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	

Frame Category	Type Value	Purpose	Example Evidence
Control	1	Flow control and acknowledgements	
Data	2	Carries network payload	

ANSWER:

```

Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwva
  Metasploitable 2
  Kali Linux

(collins@kali)-[~/SBT-DF203-Lab9]
$ CAP=working/file_working

(collins@kali)-[~/SBT-DF203-Lab9]
$ capinfos "$CAP" | tee reports/capinfos.txt
File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 08:51:54.364525
Latest packet time:  2015-03-06 08:56:27.334320
Data byte rate:      49 kbps
Data bit rate:       395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:              c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:                cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:    False
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 45169

(collins@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

Protocol Hierarchy Statistics
Filter:

frame                                frames:45169 bytes:13511274
  wlan                               frames:45169 bytes:13511274
    data                             frames:15713 bytes:13104128

(collins@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan' -T fields \
-e frame.number \
-e frame.time \
-e wlan.fc.type \
-e wlan.fc.subtype \
-e wlan.sa \
-e wlan.da \
-e wlan.bssid | head -n 100 | tee reports/wlan_frame_sample.tsv
1      2015-03-06T08:51:54.364526000+0100      1      11
2      2015-03-06T08:51:54.364527000+0100      1      11
3      2015-03-06T08:51:54.364525000+0100      1      11
4      2015-03-06T08:51:54.364526000+0100      1      11
5      2015-03-06T08:51:54.365550000+0100      1      11
6      2015-03-06T08:51:54.366063000+0100      1      11
7      2015-03-06T08:51:54.366062000+0100      1      11
8      2015-03-06T08:51:54.366061000+0100      1      11
9      2015-03-06T08:51:54.366063000+0100      1      11
10     2015-03-06T08:51:54.366061000+0100      1      11
11     2015-03-06T08:51:54.366064000+0100      1      11
12     2015-03-06T08:51:54.366063000+0100      1      11
13     2015-03-06T08:51:54.366063000+0100      1      11
14     2015-03-06T08:51:54.366063000+0100      1      11
15     2015-03-06T08:51:54.366064000+0100      1      11
16     2015-03-06T08:51:54.366062000+0100      1      11
17     2015-03-06T08:51:54.366575000+0100      1      11
18     2015-03-06T08:51:54.366575000+0100      1      11

```

```
19 2015-03-06T08:51:54.366573000+0100 1 11
20 2015-03-06T08:51:54.366574000+0100 1 11
21 2015-03-06T08:51:54.366575000+0100 1 11
22 2015-03-06T08:51:54.366575000+0100 1 11
23 2015-03-06T08:51:54.366575000+0100 1 11
24 2015-03-06T08:51:54.366574000+0100 1 11
25 2015-03-06T08:51:54.366574000+0100 1 11
26 2015-03-06T08:51:54.366574000+0100 1 11
27 2015-03-06T08:51:54.366574000+0100 1 11
28 2015-03-06T08:51:54.366574000+0100 1 11
29 2015-03-06T08:51:54.366574000+0100 1 11
30 2015-03-06T08:51:54.366575000+0100 1 11
31 2015-03-06T08:51:54.366573000+0100 1 11
32 2015-03-06T08:51:54.366575000+0100 1 11
33 2015-03-06T08:51:54.366574000+0100 1 11
34 2015-03-06T08:51:54.366575000+0100 1 11
35 2015-03-06T08:51:54.366575000+0100 1 11
36 2015-03-06T08:51:54.366574000+0100 1 11
37 2015-03-06T08:51:54.366575000+0100 1 11
38 2015-03-06T08:51:54.366576000+0100 1 11
39 2015-03-06T08:51:54.366575000+0100 1 11
40 2015-03-06T08:51:54.366574000+0100 1 11
41 2015-03-06T08:51:54.366573000+0100 1 11
42 2015-03-06T08:51:54.366573000+0100 1 11
43 2015-03-06T08:51:54.366574000+0100 1 11
44 2015-03-06T08:51:54.367085000+0100 1 11
45 2015-03-06T08:51:54.367086000+0100 1 11

46 2015-03-06T08:51:54.367087000+0100 1 11
47 2015-03-06T08:51:54.367089000+0100 1 13
48 2015-03-06T08:51:54.367085000+0100 1 11
49 2015-03-06T08:51:54.367087000+0100 1 11
50 2015-03-06T08:51:54.367087000+0100 1 11
51 2015-03-06T08:51:54.367087000+0100 1 11
52 2015-03-06T08:51:54.367087000+0100 1 11
53 2015-03-06T08:51:54.367086000+0100 1 11
54 2015-03-06T08:51:54.367633000+0100 0 8 00:26:66:55:97:d6 ff:ff:ff:ff:ff:ff 00:26:66:55:97
:d6
55 2015-03-06T08:51:54.368111000+0100 1 11
56 2015-03-06T08:51:54.368111000+0100 1 11
57 2015-03-06T08:51:54.368110000+0100 1 11
58 2015-03-06T08:51:54.368110000+0100 1 11
59 2015-03-06T08:51:54.368110000+0100 1 11
60 2015-03-06T08:51:54.368110000+0100 1 11
61 2015-03-06T08:51:54.368622000+0100 1 11
62 2015-03-06T08:51:54.368623000+0100 1 11
63 2015-03-06T08:51:54.368621000+0100 1 11
64 2015-03-06T08:51:54.368622000+0100 1 11
65 2015-03-06T08:51:54.368621000+0100 1 11
66 2015-03-06T08:51:54.368623000+0100 1 11
67 2015-03-06T08:51:54.368622000+0100 1 11
68 2015-03-06T08:51:54.368623000+0100 1 11
69 2015-03-06T08:51:54.368622000+0100 1 11
70 2015-03-06T08:51:54.368622000+0100 1 11
71 2015-03-06T08:51:54.368623000+0100 1 11
72 2015-03-06T08:51:54.368621000+0100 1 11
73 2015-03-06T08:51:54.368622000+0100 1 11
74 2015-03-06T08:51:54.369134000+0100 1 11
75 2015-03-06T08:51:54.369134000+0100 1 11
76 2015-03-06T08:51:54.369135000+0100 1 11
77 2015-03-06T08:51:54.369648000+0100 1 11
78 2015-03-06T08:51:54.369646000+0100 1 11
79 2015-03-06T08:51:54.370159000+0100 1 11
80 2015-03-06T08:51:54.371182000+0100 1 11
81 2015-03-06T08:51:54.371181000+0100 1 11
82 2015-03-06T08:51:54.371183000+0100 1 11
83 2015-03-06T08:51:54.371694000+0100 1 11
84 2015-03-06T08:51:54.371694000+0100 1 11
85 2015-03-06T08:51:54.372207000+0100 1 11
86 2015-03-06T08:51:54.372718000+0100 1 11
87 2015-03-06T08:51:54.375279000+0100 1 11
88 2015-03-06T08:51:54.375791000+0100 1 11
89 2015-03-06T08:51:54.376813000+0100 1 11
90 2015-03-06T08:51:54.376815000+0100 1 11
91 2015-03-06T08:51:54.376815000+0100 1 11
92 2015-03-06T08:51:54.376815000+0100 1 11
93 2015-03-06T08:51:54.376814000+0100 1 11
94 2015-03-06T08:51:54.376815000+0100 1 11
95 2015-03-06T08:51:54.376814000+0100 1 11
96 2015-03-06T08:51:54.381421000+0100 1 11
97 2015-03-06T08:51:54.381423000+0100 1 11
98 2015-03-06T08:51:54.382446000+0100 1 11
99 2015-03-06T08:51:54.382446000+0100 1 11
100 2015-03-06T08:51:54.382957000+0100 1 11
```

9. Part B - Identify WEP Protection and Key Parameters

- 1. Confirm that protected data frames are present.
- 2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
- 3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.
- 4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
```

```
| head -n 200 | tee reports/wep_protected_frames.tsv
```

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \  
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

ANSWER:

Kali Linux - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

- Owasp bwwa
- Metasploitable 2
- Kali Linux

Session Actions Edit View Help

collins@kali: ~ /SBT-DF203-Lab9

```
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number \
-e frame.time_epoch \
-e wlan.sa \
-e wlan.da \
-e wlan.bssid \
-e wlan.wep.iv \
-e wlan.wep.key | head -n 200 | tee reports/wep_protected_frames.tsv
```

9432	1425628319.232982000	00:26:66:55:97:d4	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc67ffec	0
10130	1425628320.138252000	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	00:26:66:55:97:d6	0xc2cbeb	0
10155	1425628320.176656000	f0:f6:1c:68:96:7c	04:8d:38:48:e5:b4	00:26:66:55:97:d6	0xc3cbeb	0
10157	1425628320.177168000	f0:f6:1c:68:96:7c	04:8d:38:48:a8:b5	00:26:66:55:97:d6	0xc4cbeb	0
10159	1425628320.177168000	f0:f6:1c:68:96:7c	08:10:77:92:7c:2f	00:26:66:55:97:d6	0xc5cbeb	0
10161	1425628320.177168000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xc6cbeb	0
10163	1425628320.177168000	f0:f6:1c:68:96:7c	c8:3a:35:58:c4:ba	00:26:66:55:97:d6	0xc7cbeb	0
10165	1425628320.177167000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc8cbeb	0
10167	1425628320.177686000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9cbeb	0
10170	1425628320.180760000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc0ffc	0
10171	1425628320.182288000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6	0xc1cbeb	0
10172	1425628320.187417000	00:26:66:55:97:d4	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc2ffc	0
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11342 1425628339.605713000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x0bccbe 0
11344 1425628339.612882000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x0ccbe 0
11346 1425628339.612881000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x0dccb 0
11352 1425628339.786966000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xadffec 0
11354 1425628339.786966000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xaeffec 0
11362 1425628340.092182000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xafffec 0
11363 1425628340.092181000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb0ffec 0
11365 1425628340.092693000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb1ffec 0
11368 1425628340.093206000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb3ffec 0
11370 1425628340.093717000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb4ffec 0
11372 1425628340.093717000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb5ffec 0
11374 1425628340.094229000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb6ffec 0
11376 1425628340.095253000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb7ffec 0
11378 1425628340.095253000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb8ffec 0
11380 1425628340.118802000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x0ccbe 0
11382 1425628340.118802000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x0fcbbe 0
11384 1425628340.119314000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x10ccbe 0
11386 1425628340.119314000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x11ccbe 0
11388 1425628340.119314000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x12ccbe 0
11390 1425628340.120338000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x13ccbe 0
11392 1425628340.122898000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x14ccbe 0
11394 1425628340.219154000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x15ccbe 0
11396 1425628340.303637000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xb9ffec 0
11398 1425628340.307218000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x16ccbe 0
11400 1425628340.456725000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xbaffec 0
11402 1425628340.482835000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x17ccbe 0
11404 1425628340.483347000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x18ccbe 0
11406 1425628340.490803000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x19ccbe 0
11408 1425628340.491538000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x1accbe 0
11410 1425628340.559638000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xbbffec 0
11412 1425628340.562706000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x1bccbe 0
11414 1425628340.655382000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xbcffec 0
11416 1425628340.655382000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xbdffec 0
11418 1425628340.656405000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xbeffec 0
11420 1425628340.656405000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xbfffec 0
11422 1425628340.656917000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc0ffec 0
11424 1425628340.656917000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc1ffec 0
11426 1425628340.657430000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc2ffec 0
11428 1425628340.657942000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc3ffec 0
11430 1425628340.657942000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc4ffec 0
11432 1425628340.658454000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc5ffec 0
11434 1425628340.658962000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc6ffec 0
11436 1425628340.658962000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc7ffec 0
11438 1425628340.658962000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc8ffec 0
11440 1425628340.662034000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc9ffec 0
11442 1425628348.750882553 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc0ffec 0
11446 1425628340.740882000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x21ccbe 0
11448 1425628340.747538000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0x22ccbe 0
11450 1425628340.818198000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc7ffec 0
11452 1425628340.818198000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc8ffec 0
11454 1425628340.818198000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xc9ffec 0
11456 1425628340.818709000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcaffec 0
11458 1425628340.819222000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcbffec 0
11460 1425628340.819222000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcdffec 0
11462 1425628340.819734000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcdffec 0
11464 1425628340.819734000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcdffec 0
11466 1425628340.820243000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc3ccbe 0
11468 1425628340.821270000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0xcfffec 0
```

```
(collins@kali) [~/SBT-DF203-Lab9]
shark -F -i wlan.fc.protected=1 -T fields -e wlan.wep.iv | sort | uniq -c | sort -nr | head -n 30 | tee reports
/repeated_iv_summary.txt
```

```
7 0x42e8eb
7 0x28d3eb
6 0x95dfeb
5 0x9415ed
5 0x86e5eb
5 0x7ae5eb
5 0x28e7eb
5 0x19e5eb
4 0xd5d7eb
4 0xd3e8eb
4 0xc915ed
4 0xc7d8eb
4 0xad4eab
4 0x73e5eb
4 0x65d8eb
4 0x4ad5eb
4 0x31e7eb
4 0x2723ed
4 0x10e7eb
3 0xffe5eb
3 0xf609ed
3 0xa307ed
3 0x97dbeb
3 0x96d1eb
3 0x92e8eb
3 0x91e8eb
3 0x8adeeb
3 0x630aed
3 0x56e8eb
3 0x3c1aed
```

```
(collins@kali) [~/SBT-DF203-Lab9]
```

10. Part C - Recover or Validate the WEP40 Key

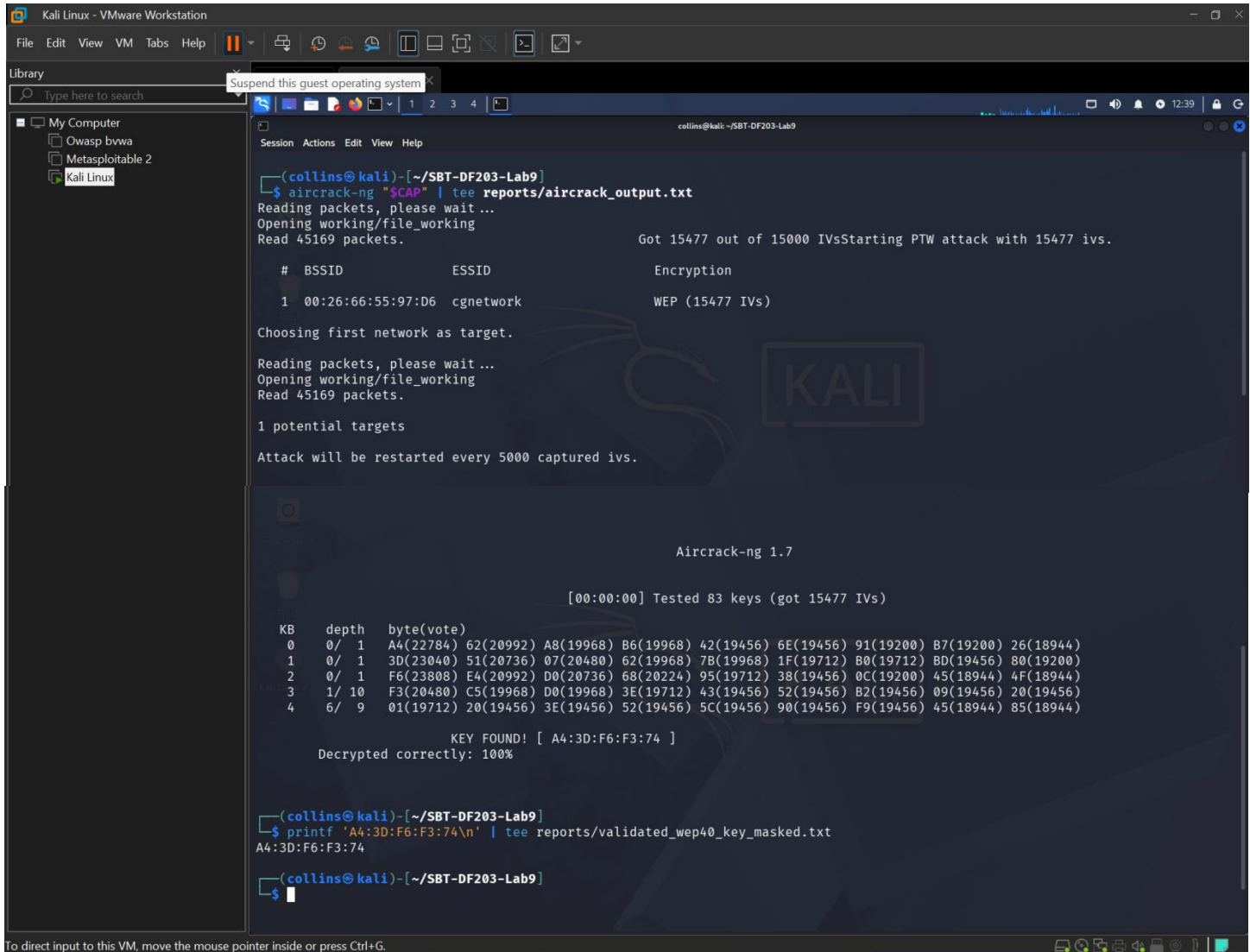
Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
```

```
# Expected historical training key from the slide: A4:3D:F6:F3:74
```

```
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```

ANSWER:



```
(collins@kali)-[~/SBT-DF203-Lab9]
$ aircrack-ng "$CAP" | tee reports/aircrack_output.txt
Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.
Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

# BSSID      ESSID      Encryption
1 00:26:66:55:97:D6  cgnetwork  WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0  0/ 1  A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944)
1  0/ 1  3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) BD(19456) 80(19200)
2  0/ 1  F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944)
3  1/ 10 F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

(collins@kali)-[~/SBT-DF203-Lab9]
$ printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

(collins@kali)-[~/SBT-DF203-Lab9]
$
```

Evidence Handling

The recovered key is part of the historical CTF dataset, not authorization to attack any wireless network. Keep it inside the course submission.

11. Part D - Decrypt the Capture Offline

```
cd working
```

```
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
```

```
cd ..
```

```
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt
```

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

ANSWER:

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwaa
  Metasploitable 2
  Kali Linux

Session Actions Edit View Help
(collins@kali) - [~/SBT-DF203-Lab9]
$ cd working

(collins@kali) - [~/SBT-DF203-Lab9/working]
$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(collins@kali) - [~/SBT-DF203-Lab9/working]
$ cd ..

(collins@kali) - [~/SBT-DF203-Lab9]
$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
3935021 12488 -rw-rw-r-- 1 collins collins 12784501 Sep 23 12:41 working/file_working-dec
3934990 13904 -rw-rw-r-- 1 collins collins 14234002 Sep 23 11:36 working/file_working
3934989 12540 -rw-rw-r-- 1 collins collins 12837660 Sep 23 11:36 working/file_working.xz

(collins@kali) - [~/SBT-DF203-Lab9]
$ sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c26977f9048227deb89882caf3c3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144cae34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

12. Part E - Extract Endpoints and Conversations

```
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
```

```
tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
```

```
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
```

```
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
```

```
tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```

ANSWER:

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwaa
  Metasploitable 2
  Kali Linux

Session Actions Edit View Help
(collins@kali) - [~/SBT-DF203-Lab9]
$ DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)

(collins@kali) - [~/SBT-DF203-Lab9]
$ echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
Decrypted capture: working/file_working-dec

(collins@kali) - [~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

Ethernet Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c 15407 12 MB 7076 616 KB 8331 11 MB
EFMNetwoks_55:97:d4 15020 12 MB 8178 11 MB 6842 593 KB
ASUSTekCOMPU_2a:c2:7a 358 281 KB 223 270 KB 135 11 KB
IPv4mcast_fb 42 6,648 bytes 0 0 bytes 42 6,648 bytes
IPv6mcast_fb 36 7,022 bytes 0 0 bytes 36 7,022 bytes
Broadcast 31 3,386 bytes 0 0 bytes 31 3,386 bytes
IPv4mcast_7f:ff:fa 31 11 KB 0 0 bytes 31 11 KB
IPv6mcast_16 10 920 bytes 0 0 bytes 10 920 bytes
IPv6mcast_02 9 630 bytes 0 0 bytes 9 630 bytes
IPv4mcast_16 3 180 bytes 0 0 bytes 3 180 bytes
IPv4mcast_02 2 92 bytes 0 0 bytes 2 92 bytes
NetisTechnol_48:e5:b4 1 42 bytes 0 0 bytes 1 42 bytes
NetisTechnol_48:a8:b5 1 42 bytes 0 0 bytes 1 42 bytes
08:10:77:92:7c:2f 1 42 bytes 0 0 bytes 1 42 bytes
TendaTechnol_58:c4:ba 1 42 bytes 0 0 bytes 1 42 bytes
IPv6mcast_ff:2a:c2:7a 1 78 bytes 0 0 bytes 1 78 bytes
```

```
173.194.127.173:80 ↔ 192.168.0.15:50887 1 54 bytes 1 66 bytes 2 120 bytes 87.51916800
0 0.0266
173.194.127.173:80 ↔ 192.168.0.15:50889 1 54 bytes 1 66 bytes 2 120 bytes 87.67737600
0 0.0261
192.168.0.15:51029 ↔ 192.168.0.9:5000 1 60 bytes 1 78 bytes 2 138 bytes 181.29902100
0 0.0005
192.168.0.15:51033 ↔ 192.168.0.9:5000 1 60 bytes 1 78 bytes 2 138 bytes 211.51609800
0 0.0056
192.168.0.15:51030 ↔ 173.194.127.231:443 0 0 bytes 1 78 bytes 1 78 bytes 210.55398400
0 0.0000
```

```
(collins@kali) [~/SBT-DF203-Lab9]
$ tshark -r "$SEC" -Y 'ip' -T fields \
-e frame.number \
-e wlan.sa \
-e wlan.da \
-e ip.src \
-e ip.dst \
-e _ws.col.Protocol | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TLSv1
43 17.172.239.31 192.168.0.15 TLSv1
44 17.172.239.31 192.168.0.15 TLSv1
45 17.172.239.31 192.168.0.15 TLSv1
47 0.0.0.0 255.255.255.255 DHCP
48 0.0.0.0 255.255.255.255 DHCP
49 192.168.0.15 17.172.239.31 TCP
50 192.168.0.15 17.172.239.31 TCP
51 192.168.0.1 192.168.0.15 DHCP
52 192.168.0.15 17.172.239.31 TLSv1
55 17.172.239.31 192.168.0.15 TCP
56 192.168.0.15 17.172.239.31 TLSv1
57 17.172.239.31 192.168.0.15 TCP
58 17.172.239.31 192.168.0.15 TLSv1
59 192.168.0.15 17.172.239.31 TCP
60 192.168.0.15 17.172.239.31 TLSv1
61 17.172.239.31 192.168.0.15 TLSv1
65 192.168.0.15 17.172.239.31 TCP
66 192.168.0.15 17.172.239.31 TLSv1
69 17.172.239.31 192.168.0.15 TCP
70 192.168.0.15 17.172.239.31 TLSv1
74 17.172.239.31 192.168.0.15 TCP
75 17.172.239.31 192.168.0.15 TLSv1
78 192.168.0.15 17.172.239.31 TCP
81 192.168.0.15 224.0.0.251 MDNS
83 192.168.0.15 224.0.0.251 MDNS
85 192.168.0.15 224.0.0.251 MDNS
87 192.168.0.15 224.0.0.251 MDNS
91 192.168.0.15 224.0.0.251 MDNS
93 192.168.0.15 224.0.0.251 MDNS
97 192.168.0.15 8.8.8.8 DNS
98 8.8.8.8 192.168.0.15 DNS
99 192.168.0.15 198.41.209.137 TCP
100 198.41.209.137 192.168.0.15 TCP
101 192.168.0.15 198.41.209.137 TCP
102 192.168.0.15 198.41.209.137 HTTP
103 198.41.209.137 192.168.0.15 TCP
104 192.168.0.15 224.0.0.251 MDNS
106 192.168.0.15 224.0.0.251 MDNS
107 198.41.209.137 192.168.0.15 HTTP
108 198.41.209.137 192.168.0.15 TCP
109 198.41.209.137 192.168.0.15 HTTP
110 198.41.209.137 192.168.0.15 TCP
111 198.41.209.137 192.168.0.15 HTTP
112 198.41.209.137 192.168.0.15 HTTP
113 198.41.209.137 192.168.0.15 HTTP
114 198.41.209.137 192.168.0.15 HTTP
115 198.41.209.137 192.168.0.15 HTTP
116 198.41.209.137 192.168.0.15 TCP
117 192.168.0.15 198.41.209.137 TCP
118 192.168.0.15 198.41.209.137 TCP
119 192.168.0.15 198.41.209.137 TCP
120 192.168.0.15 198.41.209.137 TCP
121 192.168.0.15 198.41.209.137 TCP
122 192.168.0.15 198.41.209.137 TCP
123 192.168.0.15 198.41.209.137 TCP
124 192.168.0.15 198.41.209.137 TCP
125 192.168.0.15 198.41.209.137 TCP
126 192.168.0.15 198.41.209.137 TCP
127 192.168.0.15 8.8.8.8 DNS
128 8.8.8.8 192.168.0.15 DNS
129 192.168.0.15 23.21.196.154 TCP
130 23.21.196.154 192.168.0.15 TCP
131 192.168.0.15 23.21.196.154 TCP
132 192.168.0.15 23.21.196.154 HTTP
133 192.168.0.15 23.21.196.154 TCP
134 17.172.232.98 192.168.0.15 TLSv1
135 192.168.0.15 17.172.232.98 TCP
136 192.168.0.15 8.8.8.8 DNS
137 8.8.8.8 192.168.0.15 DNS
138 192.168.0.15 199.27.79.193 TCP
139 192.168.0.15 199.27.79.193 TCP
140 192.168.0.15 199.27.79.193 TCP
141 192.168.0.15 199.27.79.193 HTTP
142 199.27.79.193 192.168.0.15 TCP
143 199.27.79.193 192.168.0.15 TCP
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160 192.168.0.15 8.8.8.8 DNS
161 8.8.8.8 192.168.0.15 DNS
162 192.168.0.15 199.27.79.193 TCP
163 199.27.79.193 192.168.0.15 TCP
164 192.168.0.15 199.27.79.193 TCP
```

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165 192.168.0.15 8.8.8.8 DNS
166 192.168.0.15 199.27.79.193 TCP
167 192.168.0.15 199.27.79.193 HTTP
168 8.8.8.8 192.168.0.15 DNS
169 192.168.0.15 31.13.70.1 TCP
170 199.27.79.193 192.168.0.15 TCP
171 199.27.79.193 192.168.0.15 TCP
172 199.27.79.193 192.168.0.15 TCP
173 199.27.79.193 192.168.0.15 TCP
174 199.27.79.193 192.168.0.15 TCP
175 199.27.79.193 192.168.0.15 TCP
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179 199.27.79.193 192.168.0.15 TCP
180 192.168.0.15 199.27.79.193 TCP
181 192.168.0.15 199.27.79.193 TCP
182 192.168.0.15 199.27.79.193 TCP
183 192.168.0.15 199.27.79.193 TCP
184 192.168.0.15 199.27.79.193 TCP
185 31.13.70.1 192.168.0.15 TCP
186 192.168.0.15 31.13.70.1 TCP
187 192.168.0.15 31.13.70.1 TLSv1
188 199.27.79.193 192.168.0.15 TCP
189 199.27.79.193 192.168.0.15 TCP
190 199.27.79.193 192.168.0.15 TCP
191 199.27.79.193 192.168.0.15 TCP
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202 199.27.79.193 192.168.0.15 TCP
203 199.27.79.193 192.168.0.15 TCP
204 199.27.79.193 192.168.0.15 TCP
205 192.168.0.15 199.27.79.193 TCP
206 192.168.0.15 199.27.79.193 TCP
207 192.168.0.15 199.27.79.193 TCP
208 192.168.0.15 199.27.79.193 TCP
209 192.168.0.15 199.27.79.193 TCP
210 192.168.0.15 199.27.79.193 TCP
211 192.168.0.15 199.27.79.193 TCP
212 31.13.70.1 192.168.0.15 TCP
213 31.13.70.1 192.168.0.15 TLSv1.2
214 192.168.0.15 31.13.70.1 TCP
215 192.168.0.15 31.13.70.1 TLSv1.2
216 192.168.0.15 31.13.70.1 TLSv1.2
217 192.168.0.15 31.13.70.1 TLSv1.2
218 199.27.79.193 192.168.0.15 TCP
219 199.27.79.193 192.168.0.15 TCP
220 192.168.0.15 199.27.79.193 TCP
221 192.168.0.15 199.27.79.193 TCP
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233 192.168.0.15 199.27.79.193 TCP
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236 31.13.70.1 192.168.0.15 TCP
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239 199.27.79.193 192.168.0.15 TCP
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248 199.27.79.193 192.168.0.15 TCP
249 199.27.79.193 192.168.0.15 TCP

(collins@kali) - [~/SBT-DF203-Lab9]
$
```

13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.

- ```
mkdir -p exported/http_objects exported/foremost
HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
```

The image shows a Kali Linux virtual machine interface. On the left is a sidebar with a 'Library' section containing 'My Computer' and sub-items 'Owasp bwva', 'Metasploitable 2', and 'Kali Linux'. The main area is a terminal window titled 'Kali Linux' with a session identifier 'collins@kali: ~ /SBT-DF203-Lab9'. The terminal shows a series of commands and their outputs: 1. 'mkdir -p exported/http\_objects exported/foremost' is executed successfully. 2. 'tshark -r "\$DEC" --export-objects http,exported/http\_objects 2>&1 | tee reports/http\_export\_log.txt' is run, resulting in an error: 'tshark: The file "/home/collins/SBT-DF203-Lab9/working/your\_capture.pcapng" doesn't exist.' 3. 'foremost -i "\$DEC" -o exported/foremost | tee reports/foremost\_log.txt' is run, resulting in an error: 'ERROR: /home/collins/SBT-DF203-Lab9/exported/foremost is not empty Please specify another directory or run with -T.' 4. 'find exported -type f -exec file {} \; | tee reports/exported\_file\_types.txt' is run, producing a long list of file types for various files in the 'exported' directory, including PNG images, JPEG images, GIF images, HTML documents, and JavaScript sources. The list of files includes: exported/foremost/png/00010214.png, exported/foremost/png/00001831.png, exported/foremost/png/00001765.png, exported/foremost/png/000020736.png, exported/foremost/png/00010671.png, exported/foremost/png/00024242.png, exported/foremost/audit.txt, exported/foremost/jpg/00000104.jpg, exported/foremost/jpg/00004538.jpg, exported/foremost/gif/00001260.gif, exported/foremost/gif/00002450.gif, exported/foremost/gif/00010456.gif, exported/foremost/gif/00000406.gif, exported/foremost/gif/00003719.gif, exported/foremost/gif/00003653.gif, exported/foremost/gif/00010876.gif, exported/foremost/htm/00024427.htm, exported/foremost/htm/00001639.htm, exported/http\_objects/object6521: data, exported/http\_objects/share-email(1).png, exported/http\_objects/pd.js, exported/http\_objects/ogli5PAH(1).jpg, exported/http\_objects/blah(1).css, exported/http\_objects/upw-theme-standard.min(1).css, exported/http\_objects/analytics%3fonly=true&visitor\_id=736851146pi\_opt\_in=8campaign\_id=1564&account\_id=473826title=Decoding%20Zeus%20Disguised%20as%20an%20RTF%20File%20-%20PhishMe&url=http%3A%2F%2Fphishme.com%2Fdecoding-zeus-disguised-as-an-rtf-file%2F&refer.com%2F%2Ffnetsec: ASCII text, exported/http\_objects/dc(1).js, exported/http\_objects/v2(1): JSON text data, exported/http\_objects/figure-4.png, exported/http\_objects/ui-bg\_glass\_75\_e6e6e6\_1x400.png, exported/http\_objects/rss.png, exported/http\_objects/share-email.png, exported/http\_objects/lumbar.gif, exported/http\_objects/figure-16(1).png, exported/http\_objects/homewave-upwhite(1).png, and exported/http\_objects/object6684.application%2Fjavascript: JavaScript source, ASCII text, with very long lines (6674).

```
(collins@kali)-[~/SBT-DF203-Lab9]
```

## 14. Part G - Optional Historical Passphrase Challenge

The slide discusses deriving a seed and cracking the CTF passphrase using supplied challenge information. This is not required for the core WEP decryption outcome unless assigned. Do not adapt the code to live networks.

- Differentiate the 40-bit WEP key from the human-readable router passphrase.
- Record the known key, derived seed, stated character constraints and challenge hash.
- Run only the instructor-provided findseed.py or crackWEP40 script against the historical dataset.
- Document the recovered passphrase in a protected appendix, not in public screenshots.

## 15. WEP Security Analysis

- WEP uses RC4 with a short 24-bit IV, allowing IV reuse in busy networks.
- The same keystream reuse leaks relationships between plaintexts and enables statistical key recovery.
- WEP provides weak integrity and no modern protection against replay or key-recovery attacks.
- Recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported.
- Legacy WEP should be removed rather than strengthened with longer passphrases.

## 16. Required Findings Worksheet

| Question                    | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capture format and duration | <p>File name:        working/file_working</p> <p>File type:        Wireshark/tcpdump/... - pcap</p> <p>File encapsulation: IEEE 802.11 Wireless LAN</p> <p>File timestamp precision: microseconds (6)</p> <p>Packet size limit: file hdr: 65535 bytes</p> <p>Number of packets: 45 k</p> <p>File size:        14 MB</p> <p>Data size:        13 MB</p> <p>Capture duration: 272.969795 seconds</p> <p>Earliest packet time: 2015-03-06 08:51:54.364525</p> <p>Latest packet time: 2015-03-06 08:56:27.334320</p> <p>Data byte rate:    49 kBps</p> <p>Data bit rate:    395 kbps</p> <p>Average packet size: 299.13 bytes</p> <p>Average packet rate: 165 packets/s</p> |

| Question                      | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <p>SHA256:<br/>c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d</p> <p>SHA1:           cb929fd6e407c13639e3168b0f35f921788478d3</p> <p>Strict time order: False</p> <p>Number of interfaces in file: 1</p> <p>Interface #0 info:</p> <p style="padding-left: 40px;">Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)</p> <p style="padding-left: 40px;">Capture length = 65535</p> <p style="padding-left: 40px;">Time precision = microseconds (6)</p> <p style="padding-left: 40px;">Time ticks per second = 1000000</p> <p style="padding-left: 40px;">Number of stat entries = 0</p> <p style="padding-left: 40px;">Number of packets = 45169</p> |
| BSSID                         | <p>16522 00:26:66:55:97:d6</p> <p>3 00:17:c3:a7:29:69</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Primary station MAC addresses | <p>8213 00:26:66:55:97:d4    f0:f6:1c:68:96:7c</p> <p>6982 f0:f6:1c:68:96:7c    00:26:66:55:97:d4</p> <p>585 f0:f6:1c:68:96:7c    00:26:66:55:97:d6</p> <p>197 48:5b:39:2a:c2:7a    f0:f6:1c:68:96:7c</p> <p>146 f0:f6:1c:68:96:7c    48:5b:39:2a:c2:7a</p> <p>45 00:26:66:55:97:d6    ac:36:13:55:60:eb</p> <p>38 00:26:66:55:97:d6    c4:43:8f:ab:d6:36</p> <p>36 00:26:66:55:97:d6    04:1b:ba:21:4b:c5</p> <p>34 f0:f6:1c:68:96:7c    01:00:5e:00:00:fb</p> <p>31 00:26:66:55:97:d4    01:00:5e:7f:ff:fa</p>                                                                                                                                                                 |

| Question                      | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | 27 f0:f6:1c:68:96:7c 33:33:00:00:00:fb<br>26 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff<br>25 00:26:66:55:97:d6 b4:b6:76:13:6b:f9<br>22 00:26:66:55:97:d6 f0:f6:1c:68:96:7c<br>20 00:26:66:55:97:d6 50:b7:c3:26:5e:73<br>16 00:26:66:55:97:d6 00:21:5c:76:75:b1<br>12 48:5b:39:2a:c2:7a 01:00:5e:00:00:fb<br>9 48:5b:39:2a:c2:7a 33:33:00:00:00:fb<br>7 f0:f6:1c:68:96:7c 33:33:00:00:00:02<br>7 00:26:66:55:97:d6 f8:a9:d0:49:69:d1<br>7 00:26:66:55:97:d6 18:67:b0:a5:6a:dc<br>6 48:5b:39:2a:c2:7a 33:33:00:00:00:16<br>5 00:26:66:55:97:d6 10:f9:6f:8f:a8:aa<br>4 f0:f6:1c:68:96:7c 33:33:00:00:00:16<br>3 48:5b:39:2a:c2:7a ff:ff:ff:ff:ff:ff<br>3 48:5b:39:2a:c2:7a 33:33:00:00:00:02<br>3 48:5b:39:2a:c2:7a 01:00:5e:00:00:16<br>2 f0:f6:1c:68:96:7c 01:00:5e:00:00:02<br>2 00:26:66:55:97:d6 36:4c:d2:c5:84:3d |
| Protected frame count         | 15713                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Repeated IV evidence          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Recovered/validated WEP40 key | A4:3D:F6:F3:74                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Decrypted capture             | 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Question               | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| filename/hash          | working/file_working-dec                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Host IP and MAC        | 192.168.0.1 255.255.255.255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Server IP and MAC      | 192.168.0.15 255.255.255.255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Key protocols observed | <div>frame frames:15477 bytes:12536845</div> <div>eth frames:15477 bytes:12536845</div> <div>arp frames:60 bytes:2610</div> <div>ip frames:15361 bytes:12525585</div> <div>udp frames:159 bytes:31074</div> <div>dhcp frames:7 bytes:2890</div> <div>dns frames:81 bytes:10351</div> <div>mdns frames:40 bytes:6556</div> <div>ssdp frames:31 bytes:11277</div> <div>igmp frames:7 bytes:364</div> <div>tcp frames:15195 bytes:12494147</div> <div>tls frames:139 bytes:83267</div> <div>tls frames:14 bytes:12413</div> <div>http frames:278 bytes:183421</div> <div>data-text-lines frames:29 bytes:22925</div> <div>image-gif frames:7 bytes:3461</div> <div>json frames:7 bytes:6424</div> <div>data-text-lines frames:1 bytes:813</div> <div>media frames:28 bytes:21854</div> <div>png frames:29 bytes:21950</div> <div>image-jfif frames:1 bytes:435</div> <div>xml frames:1 bytes:313</div> |

| Question                | Finding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <div> <div>ipv6</div> <div>frames:56 bytes:8650</div> </div> <div> <div>icmpv6</div> <div>frames:20 bytes:1628</div> </div> <div> <div>udp</div> <div>frames:36 bytes:7022</div> </div> <div> <div>mdns</div> <div>frames:36 bytes:7022</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Recovered images/HTML   | <div>exported/foremost/audit.txt</div> <div>exported/foremost/gif/00000406.gif</div> <div>exported/foremost/gif/00001260.gif</div> <div>exported/foremost/gif/00002450.gif</div> <div>exported/foremost/gif/00003653.gif</div> <div>exported/foremost/gif/00003719.gif</div> <div>exported/foremost/gif/00010456.gif</div> <div>exported/foremost/gif/00010876.gif</div> <div>exported/foremost/htm/00001639.htm</div> <div>exported/foremost/htm/00024427.htm</div> <div>exported/foremost/jpg/00000104.jpg</div> <div>exported/foremost/jpg/00004538.jpg</div> <div>exported/foremost/png/00001765.png</div> <div>exported/foremost/png/00001831.png</div> <div>exported/foremost/png/00010214.png</div> <div>exported/foremost/png/00010671.png</div> <div>exported/foremost/png/00020736.png</div> <div>exported/foremost/png/00024242.png</div> |
| Protected appendix only |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Security conclusion     | WEP is obsolete and vulnerable to IV-reuse and key-recovery attacks; modern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Question | Finding                                       |
|----------|-----------------------------------------------|
|          | WPA2-AES/CCMP or WPA3 should be used instead. |